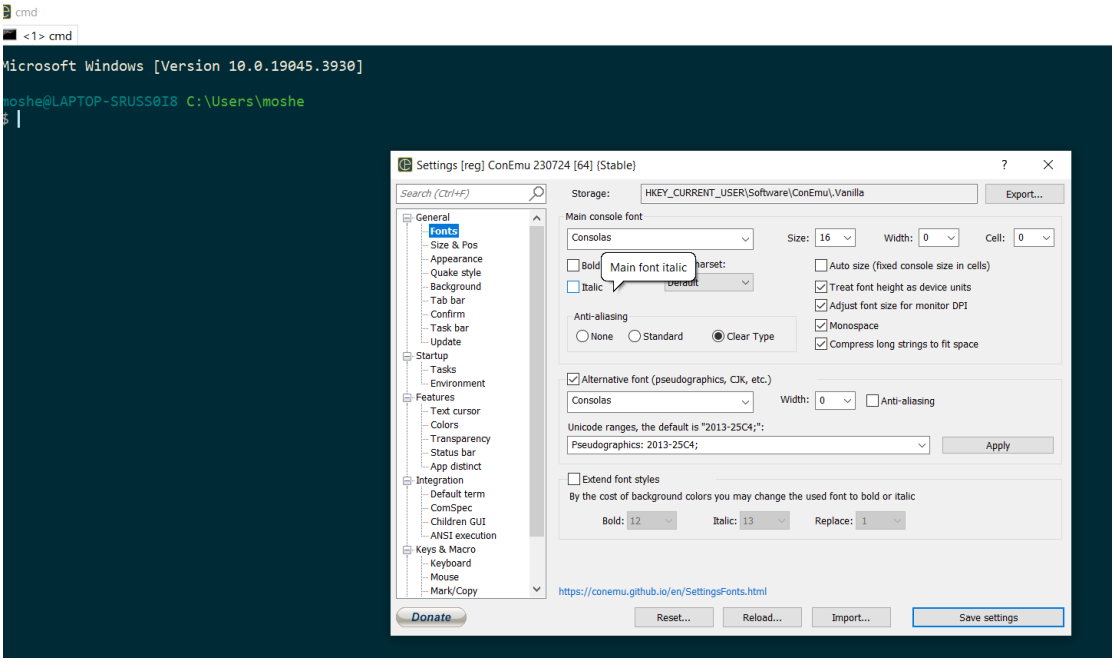
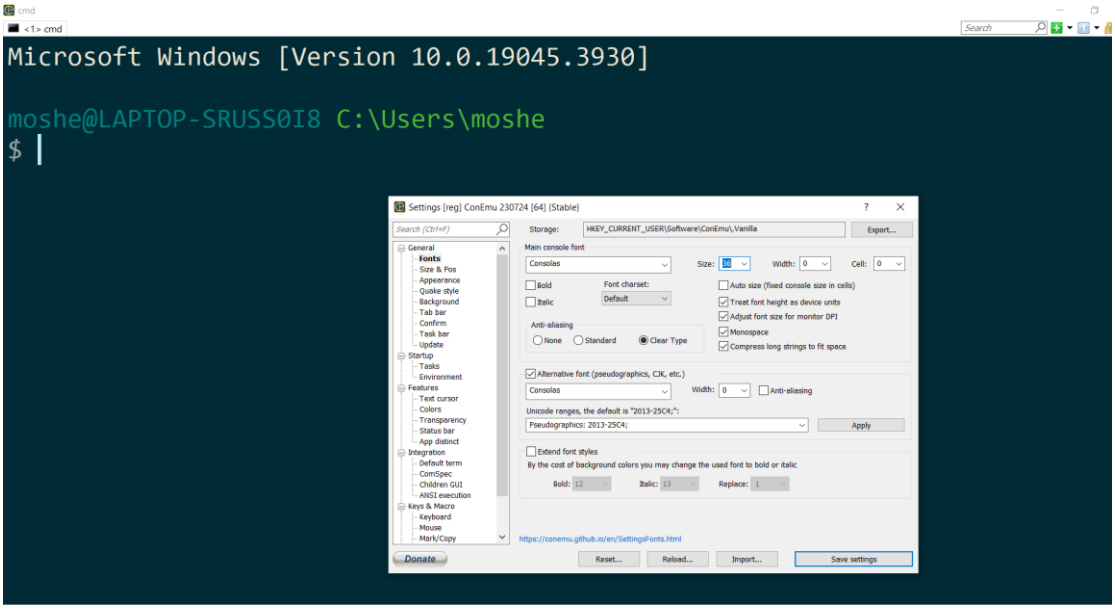
הודיה חנניאייב – 324857549

תהילה חננאייב -325435956

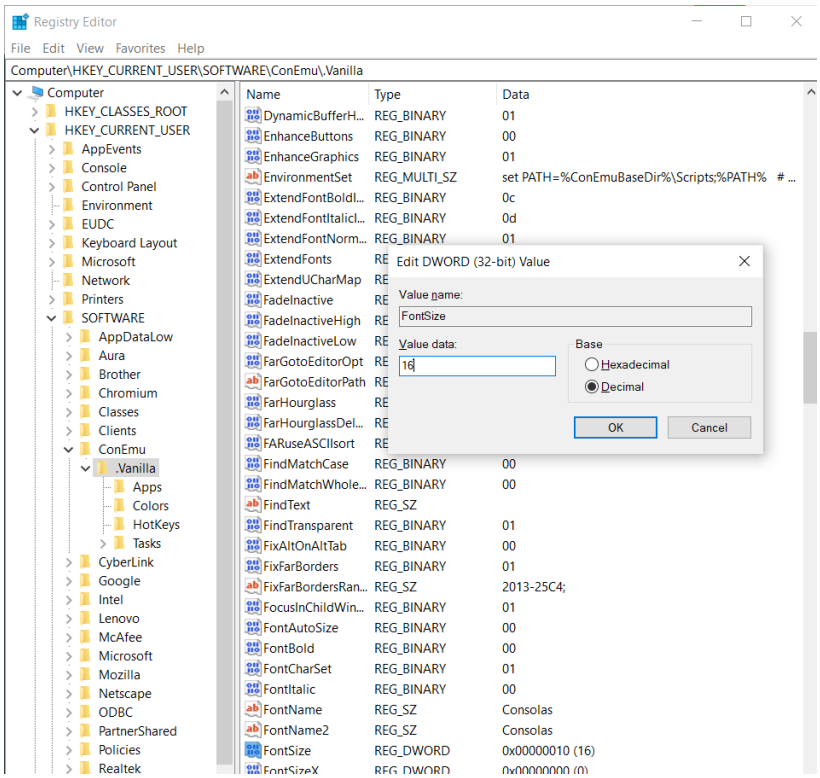
תרגיל מעבדה 1 מערכות הפעלה

.1





.2



<u>החסרונות בשמירת ההגדרות בRegistry:</u>

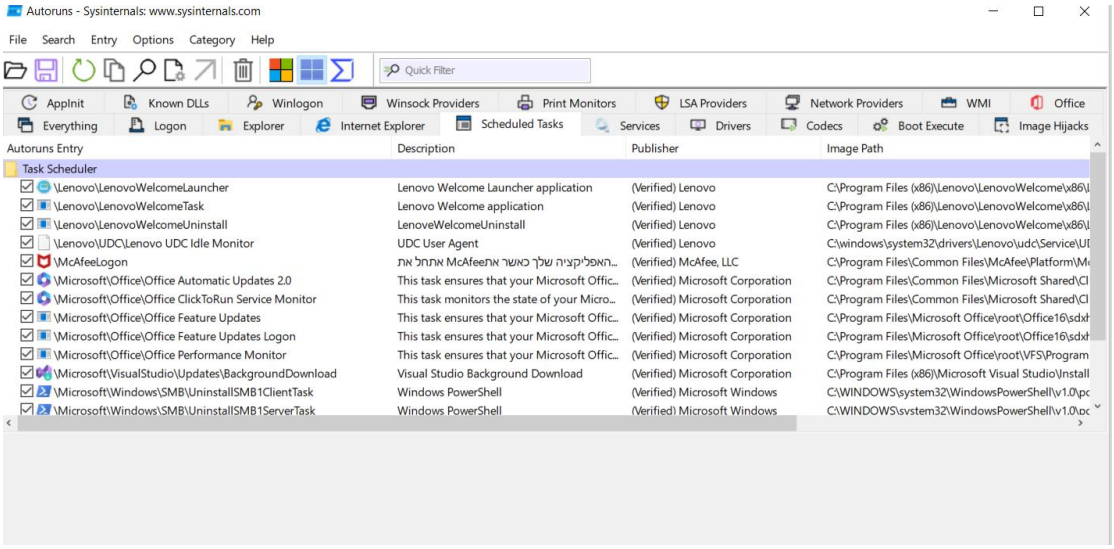
- קושי בעריכה ידנית: לעיתים קרובות יש קושי לערוך או למחוק הגדרות מרישום ה-Windows. זה עשוי להיות מסובך ומסוכן עבור משתמשים ללא התמחות בפרטי המערכת.

- סיכון של נזק למערכת: כאשר מבצעים שינויים בערכים ברישום, קיימת אפשרות של נזק למערכת אם השינויים אינם מבוצעים בצורה נכונה.

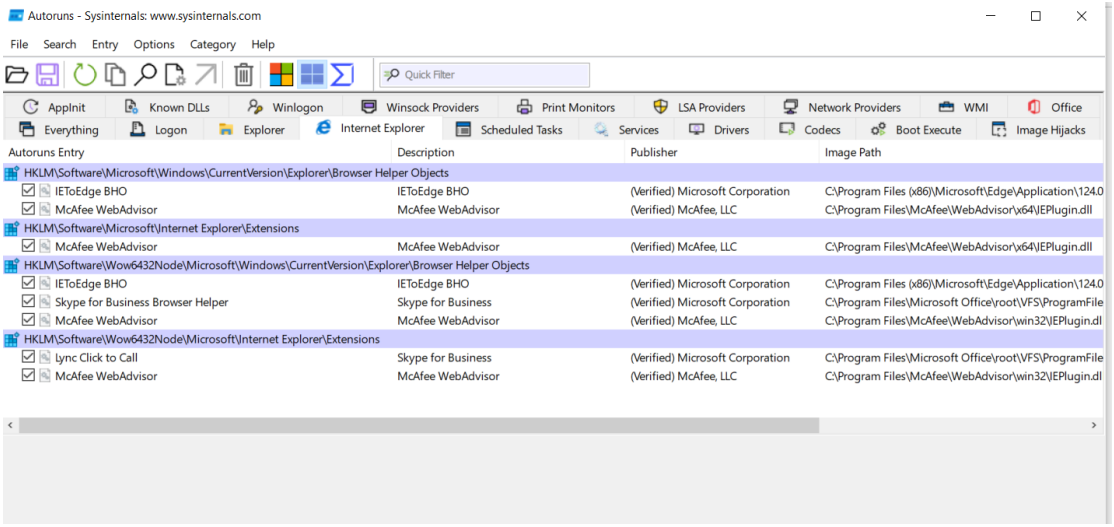
- הגבלות גישה: לפעמים יש צורך בהרשאות מנהל מערכת כדי לבצע שינויים ברישום, מה שעשוי להיות מגביל עבור משתמשים סופיים.

- אי יציבות בעדכונים: תוכנות יכולות לשנות את התצורה שלהן בעצמן גם ללא שימוש ברישום, מה שעשוי להשפיע על הגדרות שהתעדכנו ברישום.

.3



.4



.5 לא התגלו נוזקות.

.6 https://visualstudio.microsoft.com/vs/community

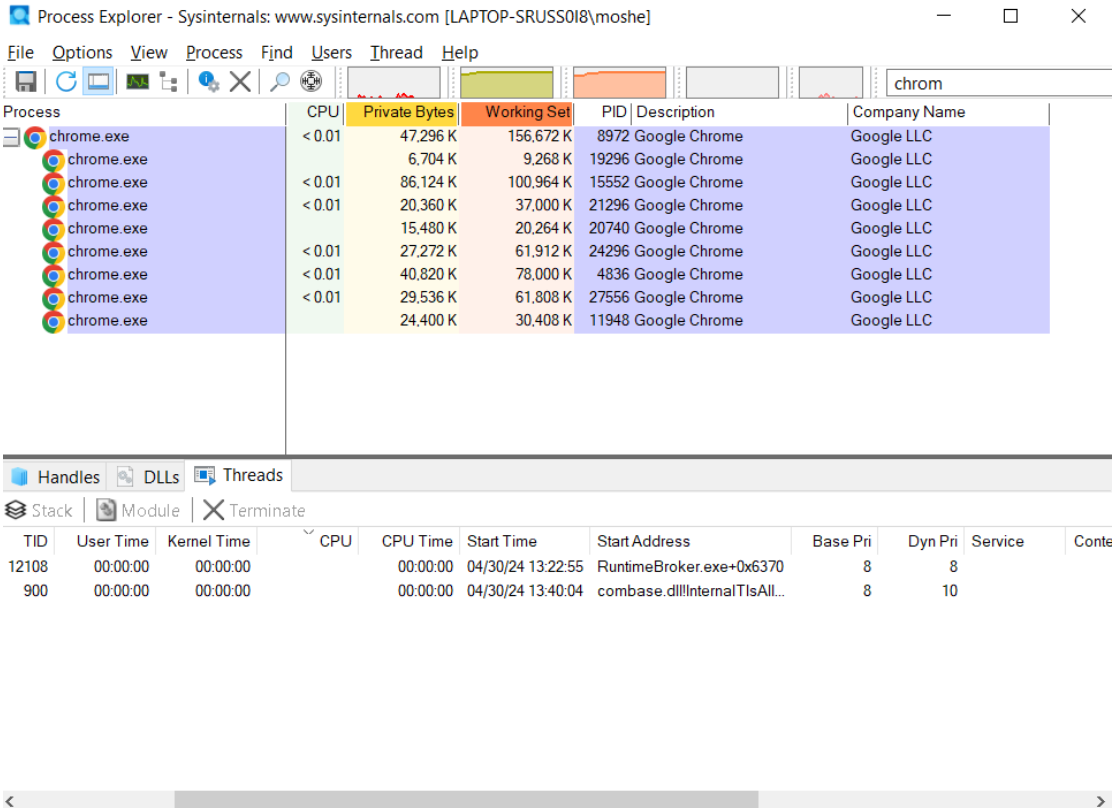
.7 שאלה 3: א. 4.88%

ב. 2

ג. 6 GB

ד. 8 GB

8. שאלה 4:



שאלה 5: א. Google LLC

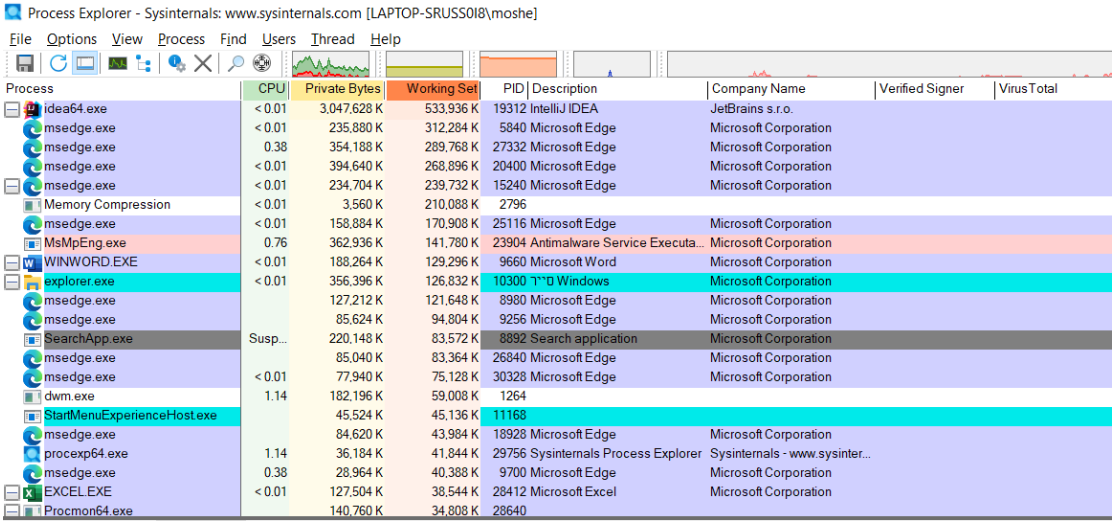
ב. 0/77

ג. אף אחד מהסטודנטים לא מצא!

ד. 72.22%

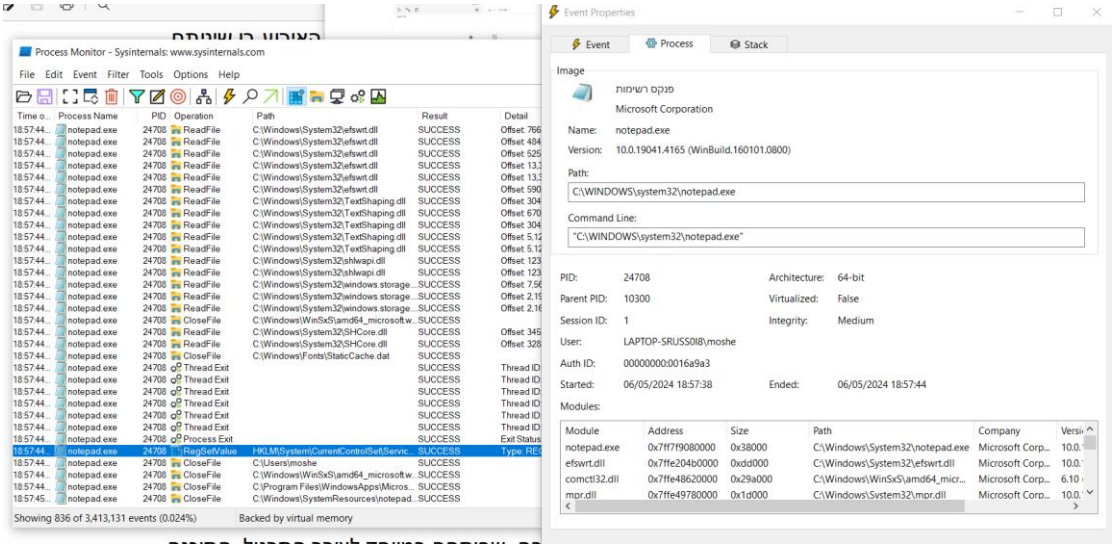
ה. 271

ו. לא, ניתן לראות בתמונה המצורפת.

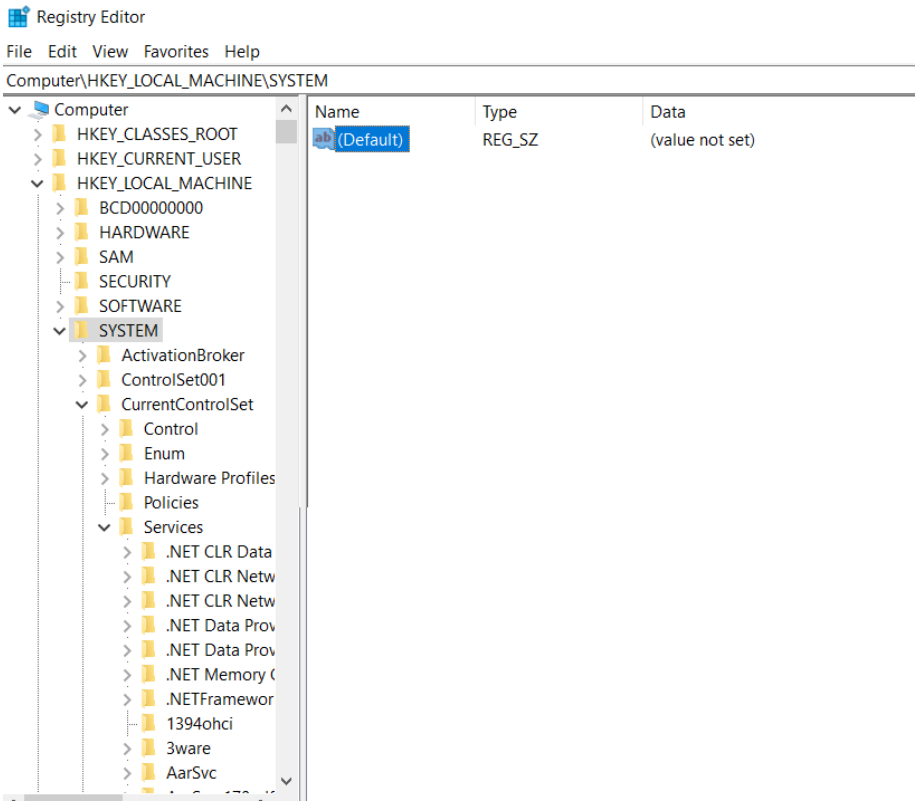


9. שאלה 7: winword

10. שאלה 3 ה':



סעיף ו':



.4

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time o...	Process Name	PID	Operation	Path	Result	Detail
19:47:13...	chrome.exe	24560	UDP Receive	LAPTOP-SRUSS0I8.Home:55907 -> tlv0...	SUCCESS	Length: 24, seqnum...
19:47:14...	chrome.exe	24560	TCP Send	LAPTOP-SRUSS0I8.Home:57358 -> 10.0...	SUCCESS	Length: 110, startim...
19:47:14...	chrome.exe	24560	TCP Receive	LAPTOP-SRUSS0I8.Home:57358 -> 10.0...	SUCCESS	Length: 110, seqnu...
19:47:15...	chrome.exe	24560	UDP Receive	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 76, seqnum...
19:47:15...	chrome.exe	24560	UDP Send	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 33, seqnum...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 221,094,400, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 221,073,920, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 221,598,208, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 221,573,632, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 223,339,008, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 223,293,952, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 223,269,376, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 223,252,992, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 223,248,896, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 217,195,008, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 222,720,512, ...
19:47:15...	chrome.exe	24560	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 221,090,304, ...
19:47:15...	chrome.exe	24560	UDP Send	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 29, seqnum...
19:47:15...	chrome.exe	24560	UDP Receive	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 25, seqnum...
19:47:15...	chrome.exe	27896	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 221,471,232, ...
19:47:15...	chrome.exe	27896	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 218,993,152, ...
19:47:15...	chrome.exe	27896	ReadFile	C:\Program Files\Google\Chrome\Appli...	SUCCESS	Offset: 219,017,728, ...
19:47:15...	chrome.exe	27896	ReadFile	C:\Users\moshe\AppData\Local\Temp\...	SUCCESS	Offset: 501,811, Len...
19:47:15...	chrome.exe	24560	TCP Disconnect	LAPTOP-SRUSS0I8.Home:57414 -> tlv0...	SUCCESS	Length: 0, seqnum: ...
19:47:15...	chrome.exe	24560	TCP Disconnect	LAPTOP-SRUSS0I8.Home:57410 -> tlv0...	SUCCESS	Length: 0, seqnum: ...
19:47:16...	chrome.exe	24560	UDP Send	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 29, seqnum...
19:47:16...	chrome.exe	24560	UDP Send	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 29, seqnum...
19:47:16...	chrome.exe	24560	UDP Receive	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 25, seqnum...
19:47:16...	chrome.exe	24560	UDP Receive	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 26, seqnum...
19:47:16...	chrome.exe	24560	UDP Send	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 29, seqnum...
19:47:16...	chrome.exe	24560	UDP Receive	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 26, seqnum...
19:47:16...	chrome.exe	24560	UDP Send	LAPTOP-SRUSS0I8.Home:56204 -> tlv0...	SUCCESS	Length: 29, seqnum...

.5

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time o...	Process Name	PID	Operation	Path	Result	Detail
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,008, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 101,900, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 101,972, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,080, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 101,972, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 1,792,000, Le...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 23,871,488, L...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 1,792,000, Le...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 1,794,560, Le...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 101,972, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 1,795,072, Le...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,116, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 101,972, Len...
9:47:13...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,116, Len...
9:47:13...	msedge.exe	15240	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 0, Length: 3,4...
9:47:13...	msedge.exe	15240	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 0, Length: 4,0...
9:47:14...	msedge.exe	15240	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 0, Length: 33...
9:47:14...	msedge.exe	15240	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 0, Length: 4,0...
9:47:14...	svchost.exe	2780	WriteFile	C:\Windows\prefetch\DLLHOST.EXE-00...	SUCCESS	Offset: 0, Length: 8,6...
9:47:14...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 101,900, Len...
9:47:14...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,080, Len...
9:47:14...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,044, Len...
9:47:14...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,116, Len...
9:47:14...	msedge.exe	9700	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 102,044, Len...
9:47:15...	svchost.exe	2780	WriteFile	C:\Windows\prefetch\DLLHOST.EXE-7D...	SUCCESS	Offset: 0, Length: 4,1...
9:47:15...	svchost.exe	1824	WriteFile	C:\Windows\System32\winevt\Logs\Micr...	SUCCESS	Offset: 200,704, Len...
9:47:15...	svchost.exe	1824	WriteFile	C:\Windows\System32\winevt\Logs\Micr...	SUCCESS	Offset: 234,312, Len...
9:47:15...	svchost.exe	2780	WriteFile	C:\Windows\prefetch\PROCMON64.EXE...	SUCCESS	Offset: 0, Length: 47,...
9:47:17...	svchost.exe	2780	WriteFile	C:\Windows\prefetch\PROCMON.EXE-2...	SUCCESS	Offset: 0, Length: 17,...
9:47:18...	taskhostw.exe	1820	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 241,664, Len...
9:47:18...	taskhostw.exe	1820	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 0, Length: 8,1...
9:47:18...	msedge.exe	15240	WriteFile	C:\Users\moshe\AppData\Local\Micros...	SUCCESS	Offset: 0, Length: 30...

Showing 10,644 of 1,615,538 events (0.65%) Backed by virtual memory

.6

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

Time o...	Process Name	PID	Operation	Path	Result	Detail
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKCU\SOFTWARE\Microsoft\InputPers...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKCU\SOFTWARE\Microsoft\Input\Typi...	SUCCESS	Type: REG_BINAR...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 16
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	Explorer.EXE	10300	RegQueryValue	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_DWO...
19:47:13...	Explorer.EXE	10300	RegQueryValue	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_DWO...
19:47:13...	Explorer.EXE	10300	RegQueryValue	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 16
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 16
19:47:13...	Explorer.EXE	10300	RegQueryValue	HKCU\SOFTWARE\Microsoft\Windows\...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	NAME NOT FOUND	Length: 144
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Setti...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_SZ, Le...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	NAME NOT FOUND	Length: 16
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...
19:47:13...	ctfmon.exe	3740	RegQueryValue	HKLM\SOFTWARE\Microsoft\Input\Loca...	SUCCESS	Type: REG_DWO...

שאלה 7.

Event Properties

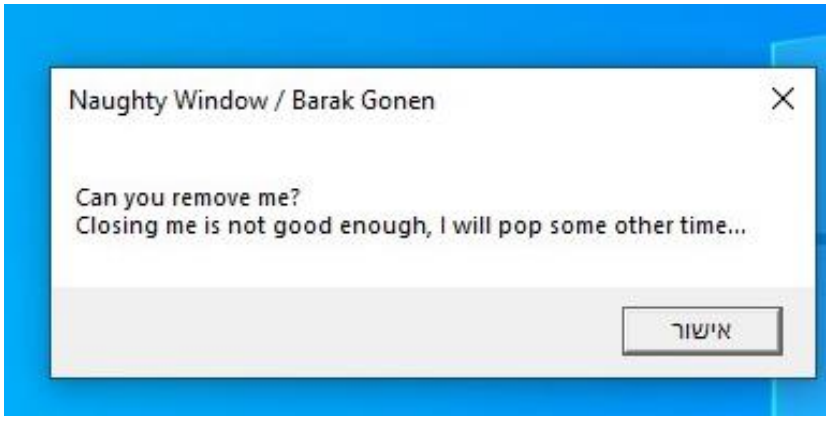
Event | Process | Stack

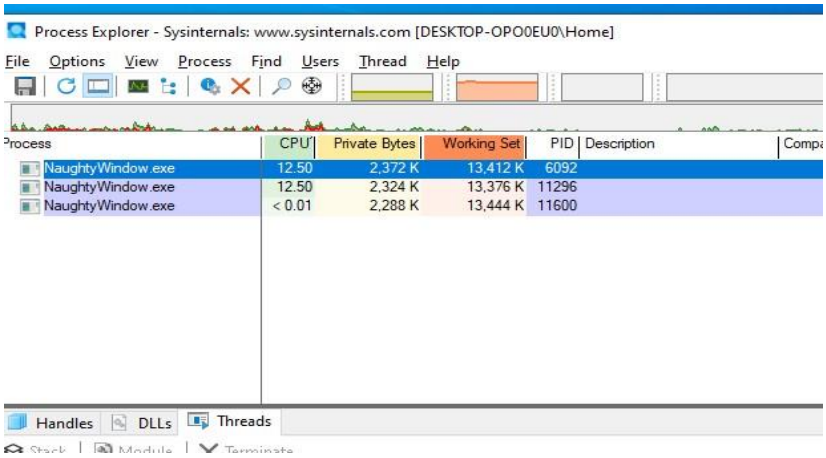
Frame	Module	Location	Address	Path
K 0	ntoskrnl.exe	ObReferenceObjectByHandle + 0x2a48	0xfffff80119048a98	C:\WINDOWS\
K 1	ntoskrnl.exe	SeUnlockSubjectContext + 0x2141	0xfffff8011904d461	C:\WINDOWS\
K 2	ntoskrnl.exe	setjmpex + 0x83f5	0xfffff80118e11235	C:\WINDOWS\
U 3	ntdll.dll	NtQueryValueKey + 0x14	0x7ffe6a08d2c4	C:\WINDOWS\
U 4	KERNELBASE.dll	MapPredefinedHandleInternal + 0x441	0x7ffe67ae3e41	C:\WINDOWS\
U 5	KERNELBASE.dll	SetProcessDynamicEnforcedCetCompatibleRanges + 0x869	0x7ffe67b643b9	C:\WINDOWS\
U 6	KERNELBASE.dll	RegQueryValueExW + 0x108	0x7ffe67ae3808	C:\WINDOWS\
U 7	KERNELBASE.dll	RegGetValueW + 0x254	0x7ffe67ae2fe4	C:\WINDOWS\
U 8	InputService.dll	InputService.dll + 0x487ab	0x7ffe2e5987ab	C:\WINDOWS\
U 9	InputService.dll	InputService.dll + 0x35f11	0x7ffe2e585f11	C:\WINDOWS\
U 10	TextInputFramework.dll	TextInputClientCreate2 + 0x3c10	0x7ffe54c03cc0	C:\WINDOWS\
U 11	TextInputFramework.dll	TextInputClientCreate2 + 0x3530	0x7ffe54c035e0	C:\WINDOWS\
U 12	TextInputFramework.dll	TextInputClientCreate2 + 0x4163	0x7ffe54c04213	C:\WINDOWS\
U 13	CoreUIComponents.dll	DllGetClassObject + 0x33ea	0x7ffe641eb1fa	C:\WINDOWS\
U 14	CoreUIComponents.dll	DllGetClassObject + 0x336e	0x7ffe641eb17e	C:\WINDOWS\
U 15	CoreUIComponents.dll	DllGetClassObject + 0xf1b8	0x7ffe641f6fc8	C:\WINDOWS\
U 16	CoreUIComponents.dll	DllGetClassObject + 0xf171	0x7ffe641f6f81	C:\WINDOWS\
U 17	CoreUIComponents.dll	DllGetClassObject + 0x1f4f9	0x7ffe64207309	C:\WINDOWS\
U 18	CoreUIComponents.dll	DllGetClassObject + 0x1f3b8	0x7ffe642071c8	C:\WINDOWS\
U 19	CoreUIComponents.dll	CoreUICreateICoreWindowFactory + 0xfa3f	0x7ffe6422ccbf	C:\WINDOWS\
U 20	CoreUIComponents.dll	DllGetClassObject + 0x1e55a	0x7ffe6420636a	C:\WINDOWS\
U 21	CoreMessaging.dll	CreateDispatcherQueueController + 0x10b3f	0x7ffe648c67df	C:\WINDOWS\
U 22	CoreMessaging.dll	CoreUICallReceive + 0x5f69	0x7ffe648f17d9	C:\WINDOWS\
U 23	CoreMessaging.dll	CreateDispatcherQueueController + 0xf1ae	0x7ffe648c4e4e	C:\WINDOWS\
U 24	CoreMessaging.dll	CreateDispatcherQueueController + 0xef15	0x7ffe648c4bb5	C:\WINDOWS\
U 25	CoreMessaging.dll	CreateDispatcherQueueController + 0xf481	0x7ffe648c5121	C:\WINDOWS\
U 26	CoreMessaging.dll	CreateDispatcherQueueController + 0xec4c	0x7ffe648c48ec	C:\WINDOWS\
U 27	CoreMessaging.dll	CreateDispatcherQueueController + 0x33e3e	0x7ffe648e9ade	C:\WINDOWS\
U 28	CoreMessaging.dll	CreateDispatcherQueueController + 0x34f42	0x7ffe648eabe2	C:\WINDOWS\

Properties... Search... Source... Save...

Next Highlighted Copy All Close

17. שאלה 4:





שאלה 7: דרך להסרת התוכנה:

- נכנסים ל process Explorer מבצעים kill process (מצאנו את התהליכים שמקפיצים את החלון בעזרת המטרה שבprocess explorer).
- ב AutoRuns מחקנו את bootstat.exe, winsafe.exe.
- הלכנו למיקום בזיכרון של כל אחד מהנוזקות לפי ה path ומחקנו את הקובץ שלהם.